

Home SOC Lab: Splunk Nmap Investigation Report

Analyst: Sree Saran S S **Date:** August 4, 2026 **SIEM:** Splunk Enterprise 10.4.2

Objective: Build a Splunk-based SOC lab, generate real attacker telemetry from Kali, and validate detection of a network reconnaissance scan against a Windows endpoint.

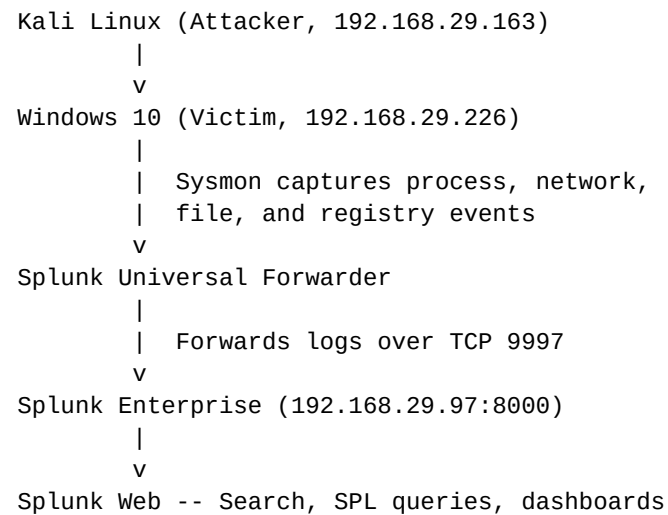
1. Lab Architecture

A three-VM home lab was built using Oracle VirtualBox to simulate a Splunk-based SOC monitoring environment. All VMs communicate over a Bridged Adapter on the local network (192.168.29.x).

Virtual Machines Deployed

VM	Role	OS	IP	Purpose
Splunk Enterprise 10.4.2	SIEM / Indexer	Ubuntu Server 22.04	192.168.29.97 (static)	Centralized log ingestion, indexing, search, and dashboard
Windows 10 Pro	Victim / Endpoint	Windows 10 (22H2)	192.168.29.226	Target endpoint monitored via Sysmon + Universal Forwarder
Kali Linux 2026.2	Attacker	Kali Linux	192.168.29.163	Offensive machine used to simulate attacks against the victim

Data Flow



2. Tools Installed & Configured

On the Splunk Server (Ubuntu)

- **Splunk Enterprise 10.4.2:** full SIEM server for indexing and searching
- **Receiving port 9997:** enabled to accept data from Universal Forwarders
- **Static IP:** pinned via netplan to keep 192.168.29.97 fixed

- **OpenSSH server**: enabled for remote file transfer (scp) from Windows host

On the Windows 10 Victim

- **Sysmon (Sysinternals)**: advanced logging for process creation (EID 1), network connections (EID 3), file creation (EID 11), registry changes (EID 13), and DNS queries (EID 22)
- **SwiftOnSecurity Sysmon config**: sysmonconfig-export.xml from GitHub for tuned event coverage
- **Splunk Universal Forwarder 10.4.2**: lightweight agent forwarding all logs to Splunk over port 9997
- **inputs.conf**: configured to forward Security, System, Application, and Sysmon Operational logs

On the Kali Linux Attacker

- **Nmap 7.99**: network scanner used for port scanning and service detection
- Pre-installed Kali penetration testing toolkit (Hydra, Metasploit, etc. staged for later attacks)

3. Pipeline Verification

Before conducting any attacks, the telemetry pipeline was verified end-to-end:

1. On Kali, ping 192.168.29.226 confirmed network reachability to Windows.
2. On Windows, the SplunkForwarder service was verified running via Services.msc.
3. On Windows CLI: `splunk list forward-server` returned:
Active forwards: 192.168.29.97:9997
4. In Splunk Web (<http://192.168.29.97:8000>), a search for `host=DESKTOP-9M1T24R` returned live events from all four log sources (Security, System, Application, Sysmon).
5. **Result**: Pipeline confirmed working - logs flow from Windows endpoint to Splunk in real time.

4. Attack Simulation: Nmap Port Scan (Reconnaissance)

Attack Description

Two Nmap scans were launched from the Kali attacker against the Windows 10 victim to simulate an adversary performing network reconnaissance. The first scan used the default SYN scan (-sS, half-open) and the second used the full TCP connect scan (-sT), allowing a direct comparison of what Sysmon does and does not detect.

Scan 1: SYN Scan (Default)

```
sudo nmap -sV -p- 192.168.29.226
```

```
(kali@kali)-[~]
$ nmap 192.168.29.226
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-04 03:17 -0400
Nmap scan report for 192.168.29.226
Host is up (0.00046s latency).
All 1000 scanned ports on 192.168.29.226 are in ignored states.
Not shown: 1000 filtered tcp ports (no-response)
MAC Address: 08:00:27:AD:6C:3B (Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 21.94 seconds

(kali@kali)-[~]
```

Figure 1: Initial Nmap scan from Kali (default SYN scan). All 1000 ports came back 'filtered' due to Windows firewall - stealthy but invisible to Sysmon EID 3.

Scan 2: TCP Connect Scan

```
sudo nmap -sT -sV -p 135,139,445,3389,7680,49668 192.168.29.226
```

```
(kali@kali)-[~]
$ sudo nmap -sT -sV -p 135,139,445,3389,7680,49668 192.168.29.226
[sudo] password for kali:
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-04 05:15 -0400
Nmap scan report for 192.168.29.226
Host is up (0.0033s latency).

PORT      STATE SERVICE      VERSION
135/tcp   open  msrpc        Microsoft Windows RPC
139/tcp   open  netbios-ssn  Microsoft Windows netbios-ssn
445/tcp   open  microsoft-ds?
3389/tcp  open  ms-wbt-server Microsoft Terminal Services
7680/tcp  open  pando-pub?
49668/tcp open  msrpc        Microsoft Windows RPC
MAC Address: 08:00:27:AD:6C:3B (Oracle VirtualBox virtual NIC)
Service Info: OS: Windows; CPE: cpe:/o:microsoft:windows

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 54.36 seconds

(kali@kali)-[~]
```

Figure 2: TCP connect scan discovered six open ports on the Windows victim, including RDP (3389) and SMB (445). Full TCP handshakes were logged by Sysmon.

Ports Discovered on Target

Port	State	Service	Version
135/tcp	open	msrpc	Microsoft Windows RPC

139/tcp	open	netbios-ssn	Microsoft Windows netbios-ssn
445/tcp	open	microsoft-ds	SMB (file sharing)
3389/tcp	open	ms-wbt-server	Microsoft Terminal Services (RDP)
7680/tcp	open	pando-pub?	Windows Update peer delivery
49668/tcp	open	msrpc	Microsoft Windows RPC (dynamic)

What Was Detected in Splunk

The Splunk timeline showed a spike of Sysmon events during the scan window, dominated by process creations and network activity:

Sysmon Event ID	Meaning	Count During Scan
1	Process Creation	99
3	Network Connection	2 (from Kali)
11	File Creation	25
13	Registry Value Set	6
Total	All Sysmon events during window	131

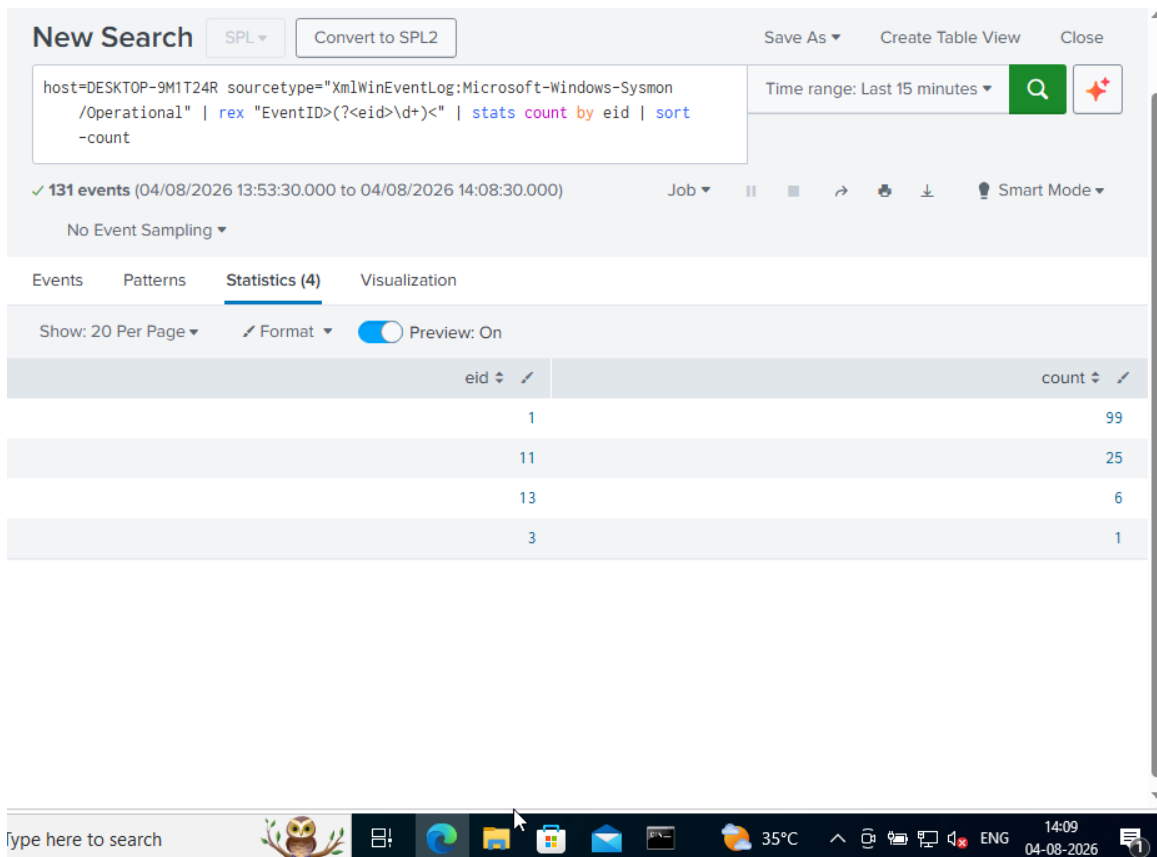


Figure 3: Splunk SPL query using rex + stats to break down 131 Sysmon events by EventID during the scan window. Process creations (EID 1) dominated the activity.

SPL Detection Query Used

The following SPL search identified attacker traffic from the Kali IP address:

```
host=DESKTOP-9M1T24R
sourcetype="XmlWinEventLog:Microsoft-Windows-Sysmon/Operational"
"192.168.29.163"
| rex "EventID>(?!<eid>\d+)<"
| stats count by eid
| sort -count
```

Query Breakdown

- host=DESKTOP-9M1T24R: filter to events from the Windows victim
- sourcetype="XmlWinEventLog:Microsoft-Windows-Sysmon/Operational": only Sysmon logs
- "192.168.29.163": raw text match for the Kali IP anywhere in the event
- | rex "EventID>(?!<eid>\d+)<": extract the numeric EventID from the raw XML into a field called eid
- | stats count by eid: aggregate and count how many of each event type occurred
- | sort -count: highest count first

Confirmed Detection Event (Sysmon EID 3)

A single expanded Sysmon Event ID 3 (network connection) captured a Kali-to-Windows RDP connection during the -sT scan. This is the strongest direct evidence of the reconnaissance:

Field	Value
Time	2026-08-04 14:45:41 IST
Host	DESKTOP-9M1T24R (Windows victim)
EventID	3 (Network Connection)
RuleName	RDP (Sysmon rule triggered)
Image	C:\Windows\System32\svchost.exe
User	NT AUTHORITY\NETWORK SERVICE
Protocol	tcp
SourceIp	192.168.29.163 (Kali attacker)
SourcePort	53418
DestinationIp	192.168.29.226 (Windows victim)
DestinationPort	3389 (RDP)
DestinationPortName	ms-wbt-server

The screenshot displays the Splunk 10.4.2 web interface. The top navigation bar includes tabs for 'Events (2)', 'Patterns', 'Statistics (0)', and 'Visualization'. Below the navigation bar, there are controls for 'Timeline format', 'Zoom Out', 'Zoom to Selection', and 'Deselect'. The main content area shows a table with columns for 'Time' and 'Event'. The 'Time' column displays '04/08/2026 14:45:41.607'. The 'Event' column contains raw XML data representing a Windows Event Log entry. On the left side, there is a sidebar with 'SELECTED FIELDS' (host 1, source 1, sourcetype 1) and 'INTERESTING FIELDS' (DestPort 1, eid 1, index 1, linecount 1, punct 1, SourceIp 1, splunk_server 1). The bottom of the interface shows a Windows taskbar with various application icons and a system clock indicating '14:58 04-08-2026'.

Time	Event
04/08/2026 14:45:41.607	<Event xmlns='http://schemas.microsoft.com/win/2004/08/events/event'><System><Provider Name='Microsoft-Windows-Sysmon' Guid='{5770385f-c22a-43e0-bf4c-06f5698ffbd9}' /><EventID>3</EventID><Version>5</Version><Level>4</Level><Task>3</Task><Opcode>0</Opcode><Keywords>0x8000000000000000</Keywords><TimeCreated SystemTime='2026-08-04T09:15:41.6074785Z' /><EventRecordID>4728</EventRecordID><Correlation><Execution ProcessID='2508' ThreadID='3604' /><Channel>Microsoft-Windows-Sysmon/Operational</Channel><Computer>DESKTOP-9MIT24R</Computer><Security UserID='S-1-5-18' /></System><EventData><Data Name='RuleName'>RDP</Data><Data Name='UtcTime'>2026-08-04 08:15:39.185</Data><Data Name='ProcessGuid'>{928f0c6c-9413-6a71-b404-000000000300}</Data><Data Name='ProcessId'>9128</Data><Data Name='Image'>C:\Windows\System32\svchost.exe</Data><Data Name='User'>NT AUTHORITY\NETWORK SERVICE</Data><Data Name='Protocol'>tcp</Data><Data Name='Initiated'>false</Data><Data Name='SourceIsIpv6'>false</Data><Data Name='SourceIp'>192.168.29.163</Data><Data Name='SourceHostname'>-</Data><Data Name='SourcePort'>53418</Data><Data Name='SourcePortName'>-</Data><Data Name='DestinationIsIpv6'>false</Data><Data Name='DestinationIp'>192.168.29.226</Data><Data Name='DestinationHostname'>DESKTOP-9MIT24R</Data><Data Name='DestinationPort'>3389</Data><Data Name='DestinationPortName'>ms-wbt-server</Data></EventData></Event>

Figure 4: Splunk expanded event view showing raw XML for the Kali-to-Windows RDP connection. SourceIp=192.168.29.163 (Kali), DestinationPort=3389 (RDP), captured by Sysmon EID 3.

Activate Windows
Go to Settings to activate Windows.

rosoft.com/win/2004/08/events/even

5. Investigation Analysis

The 6-Question Framework Applied

Question	Answer
WHO (Source)	Kali Linux attacker at 192.168.29.163
WHAT (Activity)	Nmap TCP connect and SYN scans against Windows victim, probing for open services
WHEN (Timestamp)	August 4, 2026, ~14:41-14:45 IST
WHERE (Target)	Windows 10 endpoint DESKTOP-9M1T24R at 192.168.29.226
HOW (Method)	Nmap sent TCP probes to all 65,535 ports; open ports triggered Windows service responses captured by Sysmon
WHY it matters	Reconnaissance is the first phase of the Cyber Kill Chain. Detecting it early lets a SOC block or profile the attacker before exploitation.

Key Observations

- The Splunk pipeline received 131 Sysmon events during the ~3-minute attack window, demonstrating measurable telemetry from a single Nmap scan.
- **Detection gap identified:** Sysmon Event ID 3 only logs completed TCP connections. The default Nmap SYN scan (-sS) sends SYN packets but never completes the handshake, so only 1 network connection event was captured despite thousands of probes.
- The TCP connect scan (-sT) produced 2 confirmed Sysmon EID 3 events with Kali as SourceIP, proving detection is possible when the attacker uses a full-handshake scan.
- **Analyst takeaway:** Sysmon endpoint telemetry alone is insufficient for detecting stealthy port scans. Firewall logs or Windows Filtering Platform (WFP) events are required to see filtered/dropped SYN probes.
- The parent-child process chain and DestinationPortName (ms-wbt-server) provide clear enrichment context for triage.

6. Recommended Remediations

If this were a real incident, the following actions would be appropriate:

1. Block the source IP (192.168.29.163) at the perimeter firewall.
2. Isolate the target host (DESKTOP-9M1T24R) for further investigation.
3. Review all authentication events (EventID 4624/4625) around the same timeframe for signs of successful compromise.
4. Enable host-based firewall rules to restrict unnecessary inbound connections (particularly disable exposed RDP if unused).
5. Implement network segmentation to limit lateral movement paths.
6. Configure Splunk to alert on repeated scan behavior: a single SourceIP touching >5 distinct DestinationPorts within 60 seconds.

7. Ingest Windows firewall logs to close the SYN-scan detection gap identified above.

7. Skills Demonstrated

- Deployed and configured a multi-VM SOC lab (Ubuntu Splunk server, Windows 10 endpoint, Kali attacker) from scratch on VirtualBox.
- Installed and configured Splunk Enterprise as an indexer with receiving on port 9997.
- Configured Sysmon with the SwiftOnSecurity community config for detailed Windows telemetry.
- Deployed the Splunk Universal Forwarder with a custom inputs.conf to forward Security, System, Application, and Sysmon logs.
- Executed reconnaissance scans (Nmap SYN and TCP connect) against a target endpoint.
- Wrote Splunk SPL queries using rex, stats, and sort to hunt attacker activity from raw Sysmon XML.
- Investigated detection results and identified a real telemetry gap (SYN-scan invisibility to Sysmon EID 3).
- Applied a structured 6-Question investigation methodology.
- Documented findings in a professional incident report format.

8. Next Steps

- Conduct SMB brute-force simulation using Hydra and detect via Windows Security Event ID 4625 (failed logon) floods.
- Conduct RDP brute-force simulation and correlate 4625 events with LogonType 10.
- Simulate a Metasploit payload execution and detect via Sysmon EID 1 (process creation) and suspicious parent-child chains.
- Build a consolidated Splunk dashboard with panels for each attack type.
- Publish the full lab and SPL queries to GitHub as a portfolio piece.

Report prepared by Sree Saran S S as part of SOC Analyst (L1) training and portfolio development.